

Group 6 Research Project

Final Report

WILLOW



Faculty of Science

Group members:

Youssef Baccouche
James Coenraad
Lieve Jilesen
Thijs Grootjans
Vladimir Vučković

Course:

Introduction to Artificial Intelligence

Lecturer:

Stefan Schlobach

Mentor:

Michel Klein

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1 Introduction

When we perceive a situation to be dangerous, anxiety might prevent us from engaging ourselves into that situation. Anxiety might be part of our survival instinct, but in some cases, people feel anxious in situations that are not necessary for their survival. For some people it may even be the case that their anxiety impacts their daily lives in such a way that disrupts it, and when this happens on very often, or even on a daily basis, it could be that they have an anxiety disorder. However, we have noticed that research about anxiety and generalized anxiety disorder (GAD) often focuses on adults (over 18 years of age), instead of adolescents (12-18 years) as a target group. This report will summarize our efforts to create a conceptual idea for an app which helps anxious adolescents throughout their therapy.

These anxious adolescents will have been diagnosed with GAD, as this is one of the most common anxiety disorders. We decided that to be able to measure the effects of the app and the patient's progress, we must define the symptoms of GAD and how to measure these symptoms, leading us to our first research question: *“What are the symptoms of GAD and how can we measure these symptoms?”*.

Through our next research question, *“What are the methods that have been proven effective for treating GAD?”*, we found an effective and reliable treatment to incorporate into our app. Our research has made evident that cognitive behaviour therapy (CBT) is an effective, widely used treatment for GAD, therefore, we decided to explore the options of *“How can we incorporate cognitive behaviour therapy into our app?”*.

Once we found a method to successfully incorporate CBT into our app, we must determine how effective an app could be when treating GAD by determining *“to what extent could an app help to reduce symptoms of GAD?”*. In order for an app to reduce symptoms of GAD, it must be able to recognize when the user has ruminating thoughts, hence, our succeeding research question: *“How can an app recognize ruminating thoughts and interfere?”*. These thoughts could play a major role in the severity of the patient's disorder, and it is necessary for the app to address it in order to adequately treat the patients.

After research that focuses on GAD and CBT, we recognized the importance that the design of the app holds on the user's experience. Consequently, we had to determine the design elements that could be incorporated in the app, such as colour palettes and themes, through the research question: *“Are there certain colours or themes that have a calming effect?”*.

We think it is especially important to focus on addressing GAD in adolescents, as they are often not the main target group in research. Adolescents might have different ways of coping with their disorder compared to adults, and we feel like an app might help adolescents to seek treatment more actively because the mental barrier to download an app is likely to be lower, than to reach out to a therapist, for example. By creating an app, we want to contribute to making therapy for GAD more widely accessible, and potentially (if proven effective) decrease waiting list times by increasing the effectiveness of therapy.

2 What is the problem?

It is unclear as to what the exact cause of GAD is, yet there is a range of factors that have been proven to contribute, such as the following: biological factors, genetics factors, life experiences (possibly traumatic events). The disorder may begin during any period within the life cycle and gradually intensify, however, it usually appears between childhood and middle age.

GAD is characterized by consistent and excessive worry. We can distinguish two different kinds of symptoms: emotional and physical symptoms. Sweating, shortness of breath and increased heart rate fall under physical symptoms. While mental symptoms include excessive fear/worry and reduced ability to concentrate. These mental and physical symptoms can hinder basic cognitive processes as simple as thinking¹.

Every year, 6.8 million adults in the United States suffer from GAD, with two thirds of patients struggling with the disorder being female. Out of these 6.8 million suffering adults, only 36.9% seek treatment. This number may seem significantly low, however, in the case of adolescents with GAD, the statistics plummet with only 25% receiving treatment. This has led us to solely focus on adolescents in our research project².

The disorder tends to fly under the radar, as a significantly low percentage of patients seek treatment. It is possible that these patients are not aware of having the disorder and may ignore or wrongly perceive the symptoms. Therefore, we will need to raise awareness of the disorder and encourage people to take a GAD test.

Another obstacle met by GAD patients is the potentially high expenses of therapy. Not everybody is aware of having GAD, and even if they are, some might not be able to afford the treatment. Additionally, even if a patient is aware of having the disorder and is able to pay for treatment, they are not always likely to get it. This is due to the large demand for therapists leading to lack of availability and these patients having to settle for the waiting list.

Our main goal is to provide a free, simple and accessible potential treatment for GAD. We aim to help adolescents in developing coping mechanisms and thoughts that will be useful to provide short term comfort, as they are experiencing anxiety. Simultaneously, we would like to provide them with supportive guidance to help gradually relieve the effects of GAD in the long term. By providing our free app, any patient struggling with GAD has hope of nursing the disorder for absolutely no cost and with no waiting time. They will be able to access this treatment with the tap of a button, anywhere and at any time.

3 What has been done?

We first gathered literature and tried to find out more information that could help us with solving our research questions. Most of the literature was gathered on *Google Scholar*, while paper 3 and 4 were found on *Citeseer*. Most of our searches were connected to GAD, how to measure symptoms of GAD, CBT and how technology can help in treatment of anxiety

¹ Han, 2009

² <https://adaa.org> (Anxiety and Depression Association of America), 21/10/2020

disorders. After gathering literature, we were able to define our concept better and therefore have left out some of the papers in our report (which are still included in section 3.1). Since then we have also added several papers to be able to define our research better and improve our knowledge about GAD. We have also conducted an interview with a licensed CBT therapist to acquire more knowledge from her expertise.

3.1 Literature Overview

Paper 1: A Randomized Controlled Trial of the Cool Teens CD-ROM Computerized Program for Adolescent Anxiety (Wuthrich et al., 2012)

In this study of 43 adolescents with a primary diagnosis of anxiety either followed a twelve-week program called “Cool Teens” or were put on a waiting list. User preferences indicated that the computerized treatment was well suited to anxious adolescents. The “Cool teens” group showed to have significant reductions in the total number of anxiety disorders, the severity of the primary anxiety disorder and the average severity for all disorders.

Paper 2: Anxiety, Affect, and Activity in Teenagers: Monitoring Daily Life With Electronic Diaries (Henker et al., 2002)

This research paper shows that it is possible to track anxiety through monitoring the daily diaries. After analysing their dairies and questionnaires students were categorised into three groups: low-, middle- and high-anxiety. High-anxiety teenagers, compared with low-anxiety teenagers, expressed higher levels of not only anxiety and stress but also of anger, sadness, and fatigue, along with lower levels of happiness and well-being.

Paper 3: Measuring Anxiety in Children: A Methodological Review of the Literature (Han, 2009)

The paper defines what anxiety is and how it can be measured. We can subdivide anxiety attacks under numerous kinds: panic disorder, agoraphobia, social phobia, simple phobia, obsessive-compulsive disorder, post-traumatic stress disorder, and generalized anxiety. There are two main kinds of anxiety: state anxiety and trait anxiety.

Paper 4: The influence of preferred relaxing music on measures of state anxiety, relaxation, and physiological responses (1989) (Davis & Thaut, 1989)

The purpose of this study was to measure physiological and psychological responses to preferred, relaxing music. Physiological data showed that the music aroused and excited rather than soothed autonomic and muscular activity. Music did calm the people down, but this only worked for state anxiety.

Paper 5: Psychological treatment of depressive rumination (Watkins, 2015)

In this article, we saw that there are good theoretical and empirical reasons to hypothesize that targeting rumination may enhance treatments for depression. Recent trials of CBT that explicitly targeted rumination support this hypothesis, with large between-treatment group

effect sizes that compare favourably to existing treatments and with changes in rumination repeatedly mediating symptom improvement.

Paper 6: Rumination as a transdiagnostic factor in depression and anxiety (McLaughlin & Nolen-Hoeksema, 2011)

This experimental study shows that inducing to ruminate in response to dysphoric mood leads to increases in both depressed and anxious moods. Which can tell us that we can identify anxious behaviour by analysing or acknowledging rumination. The Ruminative Responses Scale of the RSQ includes 22 items describing responses that are self-focused (e.g., “I think, ‘Why do I react this way?’”), symptom focused (e.g., “I think about how hard it is to concentrate”), and focused on the possible consequences and causes of negative mood.

Paper 7: Online Gaming Addiction: The Role of Sensation Seeking, Self-Control, Neuroticism, Aggression, State Anxiety, and Trait Anxiety (Mehroof, n.d.)

Trait and state anxiety were both significantly associated with online gaming addiction scores, suggesting that both internal and external anxiety factors encourage excessive online gaming (although state anxiety scores were more highly correlated than trait anxiety). This may be because some university students are faced with temporary unpleasant emotional arousal stemming from high workload. In order to decrease such anxieties, students may use online gaming as a coping strategy to reduce tension (i.e., as a mood modifier).

Paper 8: A randomized controlled trial of online versus clinic-based CBT for adolescent anxiety (March, n.d.)

Online delivery of CBT, with minimal clinician support is as efficacious as the same therapy content delivered on a face-to-face basis within a clinic setting in the treatment of anxiety disorders among adolescents. This approach offers a credible alternative to clinic-based therapy, with benefits of reduced therapist time, and greater accessibility for families who have difficulty accessing clinic-based CBT.

Paper 9: Color and emotion: effects of hue, saturation, and brightness (Wilms & Oberfeld, 2018)

The purpose of this study is to help determine the effect of three chromatic colours (red, blue, green) on the emotional state of 62 participants. The importance of this study in our project is that it will allow us to scientifically choose the colour scheme that we will incorporate within the app to help calm the patients. Therefore, we are searching for the colours that allow for minimal arousal.

Paper 10: Efficacy of Cognitive Behavioural Therapy (CBT) for anxious children and adolescents (Sawyer & Nunez, 2014)

The study helps determine how successful the use of cognitive behavioural therapy is in treating children from the age of 4 to 18. The research was able to conclude that individual

CBT decreased the rates of anxiety diagnoses and was as, or more, effective than other, similar therapies.

Paper 11: Molecularly selective nanoporous membrane-based wearable organic electrochemical device for non-invasive cortisol sensing (Parlak et al., 2018)

Stress plays an important role in the overall health of a patient; when under stress, the adrenal gland releases cortisol and adrenaline into the bloodstream. The cortisol levels in various bodily fluids can range from 4 pM to 70 µM depending on the fluid. In sweat, the optimum level of cortisol ranges from 0.02 to 0.5 µM.

Paper 12: Effects of Walking in a Forest on Young Women (Ikei et al., 2019)

The aim of this study was to evaluate the physiological and psychological effects of brief walks in forests on young women. Overall, 12 participants in each area (60 participants in total, mean age: 21.0 ± 1.3 years) were instructed to walk in a forest and a city area for approximately 15 min; simultaneously, their heart rate variability, heart rate, blood pressure, and pulse rate were measured to quantify their physiological responses to walking. A brief walk in a forest resulted in physiological and psychological relaxation effects in young women.

3.2 Interview with an expert

An interview was conducted with licenced therapist and psychologist Doris Rafajlovski. Beside her current private counselling practice, she has worked with patients and victims of gender discrimination, domestic violence and was in charge of a group that helped refugees. She obtained a Bachelor's and a Master's diploma at University of Belgrade, Faculty of Philosophy - department for Psychology. Her direction of psychology is connected with REBT (Rational Emotional Behaviour Therapy) and CBT (Cognitive Behaviour Therapy).

We conducted the online interview through *Discord* and first gave a short explanation of our concept and current app idea. After that we asked Doris to first provide us with a short introduction of herself, after which we were able to ask her several questions (appendix 4).

3.3 Conclusion about the interview

Doris was able to provide us with more insights to understand and approach the problem by sharing her expertise with us. According to Doris - to recognise anxiety there are a few common physical symptoms such as: headache, heartbeat, sweating and shortness of breath. There is still more research necessary to specify how the healthcare system is structured in the Netherlands or the country where the research will take place. We also should focus on where the adolescent first reports their anxiety. She explained more about how we could recognise a situation in which the user is anxious. According to her the most common anxious thoughts are “*I will die*” and “*I will go crazy*” during an anxious episode in which the patient feels as if they are in immediate danger.

Doris recommended that we focus on one type of anxiety. She gave panic attacks as an example or proposed we incorporate CBT into the app, as this therapy is already widely used,

very structured and an app could be beneficial to help the user with their therapy “homework”. Therefore, we decided to explore the option to incorporate CBT into our app further. Recommended homework techniques in CBT given by a therapist for people suffering from anxiety are usually in some kind of diary which will track their emotions and what happened when they felt anxious, which could be incorporated in the app.

Doris emphasized that our app should not forbid anxious thoughts from happening, as when you try to forbid these thoughts, the thoughts are likely to become stronger and will remain active. Instead we should try to focus on the previously mentioned techniques for stabilization (coming back to neutral thoughts), such as CBT.

4 How can the problem be solved?

4.1 Computational solution

The application will ask the user for two types of data input, the results that will be computed are based on symptoms for GAD, which include headaches, increased heart rate, sweating or shortness of breath³, but also mental symptoms, such as feeling anxious or worried⁴. The app will rely on self-reported data, with the added option to link a sensory device to the app as well to collect physiological data.

Self-reported data will come directly from the input of the user. Personal thoughts can be submitted through text input in the form of a diary or chat box, so that the app may be able to recognize a pattern by gathering data on where or when the user is often anxious. Physiological data will be derived from sensory devices (e.g. a wearable watch, ring or non-intrusive cortisol sensor device). The sensors might be able to collect heartbeats per minute, breathing patterns or speed of movement from the user⁵. However, to make sure that the user is actually anxious when physiological data is measured and a certain threshold is passed, the app will trigger a response to let the user indicate if they are actually feeling anxious, or for example have an elevated heart rate because they are jogging in the park. The system will compare the self-reported data with the physiological data to prevent a false alarm from happening. When the application receives mixed signals from physiological data and self-reported data, the AI will choose to use the self-reported data for recommendations and responses instead of physiological data.

We can use artificial intelligence to analyse the datasets by using natural language processing, reinforcement learning and deep learning. Deep learning allows for recognizing certain keywords or patterns in the self-reported data input from the user and categorizing this data into clusters. The clusters could be formed by categorizing the words into moods such as “nervous”, “uncomfortable” and “worried”. We will start training the AI to recognize such words by providing the AI with a list of words that are commonly used by anxious

³ Han, 2009

⁴ Spitzer et al., 2006

⁵ Parlak et al., 2018

adolescents. We will assign words to corresponding moods (especially anxious moods) and label them as “keywords”, we will improve and adjust this initial list of words through a first test trial (see section 5.1). To be able to analyse these words and put them into categories we could use natural language processing (NLP).

NLP will allow us to find keywords in the user input data to categorize them accordingly, this will be done by filtering the words based on how often a word is used, when these words are used (during an anxious episode or in daily activity) and the type of word. It is especially tricky to filter out a certain type of word, because the AI will not know by itself which words have more meaning than others; for example, “the” is often used in a sentence, but is unlikely to represent how anxious a user is at that moment. Accordingly, we have decided that our AI will focus on words that are adjectives and nouns, to reduce the possibility of a false alarm (especially at the beginning of the study).

By incorporating deep learning, we can reduce the frequency of false alarms over time by creating a weighted neural network. Through assigning weights to keywords, the AI is able to analyse which words will indicate a possible anxious episode. Once the user fills in a survey about the severity of their anxious episode after the episode has passed, the weights of the words used by the user during the episode can be adjusted accordingly. The more input (and ratings of severity) the app will get from the user, the more the weights can be adjusted to make better predictions and the less likely words like “the” are to trigger a response in the system. If a false alarm does happen in the system (response is triggered from the app while the user is not anxious) the weight of the word that contributed to the mistake the most will be adjusted. However, the more complex the neural network becomes (the more weight is added), the more data we will need to train the AI. This is why we have chosen to send the user of the app at least one reminder per day, to gather as much data from the user as possible.

The AI could potentially also look for a pattern for a location or time when a user is often anxious, and when the user is in a similar location (for example, school or a supermarket) or time of day (for example, at night) recommend to revise their CBT homework.

We would also like for our AI to be perceived as social and interactive. The app will have a virtual guide - with the goal to draw interest to the app and keep the attention of the user. The guide and this might lead to the application feeling more personal to the user, which can possibly help to make the user input more reliable, as the user might “trust” the app more.

4.2 Design

We have incorporated a willow tree as the logo of our app and a forest theme will be used throughout the project. The willow tree will grow more along with the user making progress in reducing their symptoms. We have chosen to integrate a natural theme as research indicates that nature can be perceived as calming⁶. Pastel colours are described as “soothing”, and the hue, saturation, and brightness are influential to the user experience⁷. The goal is to engage the user as much as possible to use the app, as well as to make the app as non-provocative as

⁶ Song et al., 2009

⁷ Wilms & Oberfeld, 2018

possible. Therefore, the colours used in our application are inspired by nature and calm nuances⁸.

5 Evaluation/Experimental setup

5.1 Experimental Setup

In order to test how effective the app is, we will set up and conduct an experiment that includes four trial groups. This will allow us to test the app when used in combination with therapy, and also determine whether the app has any sufficient benefits without the inclusion of therapy (so it might be used as a temporary replacement for therapy while in a waiting list). Therefore, through this, we would like to provide an alternative for adolescents that are unable to, or do not wish to see a therapist.

This experiment will be divided into two stages:

The first stage will be a pilot study with a small number of participants (small-N design) that are already receiving CBT over the course of a few weeks. This will allow us a first insight into user experience and enable us to improve the system and train the neural network before we test the app over a longer period of time. After the pilot study we will be able to make necessary changes to the app to increase overall effectiveness, which will then be tested in a longer stage over the course of several months. In the longer, second stage of the study we will compare the results of four trial groups, in which participants will be randomly assigned to one of the following groups:

CBT with the app	CBT without the app	Waiting list with the app	Waiting list without the app
This experimental group will follow CBT with a licensed therapist while simultaneously having access to the Willow app.	This control group will follow CBT with a licensed therapist without use of the Willow app.	This experimental group will be put on a waiting list to receive therapy while using the Willow app.	This control group will be put on a waiting list to receive therapy without the addition of the Willow app.

We plan to use a mixed factorial design: as one independent variable is manipulated as independent groups and the other is manipulated as within groups. We chose a mixed design for this study because it is unethical to only let one group receive treatment, accordingly the waiting list group will also experience the independent variable of CBT. However, since only the waiting lists group experiences two levels of the independent variable of CBT/waiting list, the initial CBT groups will continue to follow treatment until they no longer fit the diagnosis of GAD or they drop out of the treatment/study. It is also unethical to put the initial CBT groups on a waiting list while they were already receiving therapy, and might not have reduced

⁸ Wilms & Oberfeld, 2018

their symptoms yet, which is why the initial CBT groups will not be experiencing the independent variable of the waiting list.

Factorial 2 x 2 design	App	No app
CBT	<i>Results</i>	<i>Results</i>
Waiting list	<i>Results</i>	<i>Results</i>

The participants in the waiting list groups serve as a control group to research if the app also has a significant effect of reducing symptoms of GAD without regular sessions with a therapist. We wanted to include this control group because if the reduction of symptoms by using the app without therapy sessions is significant, a system could be implemented where the app is used by patients in a waiting list to receive therapy. By implementing this system, we could also potentially reduce the number of hours per patient a therapist might need on a weekly basis, which will allow for shorter waiting times as a therapist might be able to take on more patients.

The diagnosis of GAD will be assessed through the survey (appendix 3), conducted by a team of independent researchers and licensed therapists. Participants will also fill in a self-report survey to assess their symptoms. The severity of GAD symptoms in each participant will be measured and recorded at both the beginning and the end of the study (pre-test/post-test design). This is so we can make sure that we can track the reduction in symptoms, as well as being able to conclude that random assignment created truly equal groups in terms of severity in symptoms, age, and gender.

We have also indicated possible reasons why the app might not be used as much:

- The app and interaction might not feel “human” enough
- The user is not interested in using the app on a regular basis
- It is hard to concentrate on using the app when the user is anxious

If through a survey at the end of the first stage with a small-N design, it is found that one of these reasons, or others reasons are given for decreased app usage, we will adjust the app to be more attractive to use as our computational solution (see section 4.1) relies on getting frequent data from the user.

5.2 Evaluation strategy

To make sure that the app is reducing symptoms we will conduct a survey amongst users, and where possible their therapists to analyse the reduction in symptoms. The user input might also be analysed to see if the user is using less anxious and/or negative keywords compared to when they first started using the app. We will then compare the survey answers to the analysed user input, sensor data and the opinion of the user’s therapist (where applicable) to see if the user experience overlaps with a significant reduction in symptoms.

To measure the participant's anxiety levels, we will use a questionnaire⁹ (see appendix 3) consisting of 7 items. The questionnaire also indicates how often the participant felt anxious during the last two weeks and how much each symptom was bothering them. This questionnaire will be conducted by an independent, licensed therapist or researcher at the beginning and at the end of the study to measure the change in severity of symptoms.

5.3 Potential Difficulties

The measuring of symptoms at the beginning and end of the study needs to be controlled for potential observer biases. Observer bias can be controlled by using a masked design. In a masked design the participants will know which group they are in (this is hard to avoid in this study because the participants will be aware whether or not they are receiving therapy and/or using the app). However, the observers will not be aware of which group the participants are in.

We must also make sure to note any possible outliers systematically dropping out and thereby causing an attrition threat. However, this is prevented by using the pre-test/post-test design, as when outliers drop out of the study, we will have recorded their symptoms on the pre-test and will simply eliminate their data from the study.

6 Conclusion

To be able to measure anxiety we will use the GAD 7 scale developed by Robert L. Spitzer et al (Appendix 3). According to this scale, the severity of symptoms can be measured for a GAD patient. To measure the symptoms, the patient is interviewed by a psychologist in addition to completing a survey at the beginning and the end of the study (pre-test/post-test design). We will start by conducting a pilot study to improve our app before commencing with the main study, which will consist of a mixed 2×2 factorial design. The variables used in the factorial design are following CBT or being put on a waiting list, and using the app or not using the app.

Cognitive behavioural therapy has been proven effective for treating GAD and can be incorporated into the app. To be able to incorporate cognitive behavioural therapy into our app we will let the app function as a “homework” reminder when the user is anxious, after which the user is recommended to revise CBT exercises.

The AI will recognise certain keywords related to anxiety symptoms from the user input and remind the user of their CBT homework when a certain weight in the neural network is exceeded. The weight threshold might be exceeded once a keyword with a heavy weight is used, or several anxiety-related keywords are used within a time period of three hours. Once the weight threshold is passed, the app will recommend to revise CBT exercises. If the user decides to link a sensory device, a certain threshold will be implemented for the device and

⁹ Spitzer, R. et al., 2006

when this threshold is passed, it will first ask the user for text input before recommending to revise, to decrease potential false alarms.

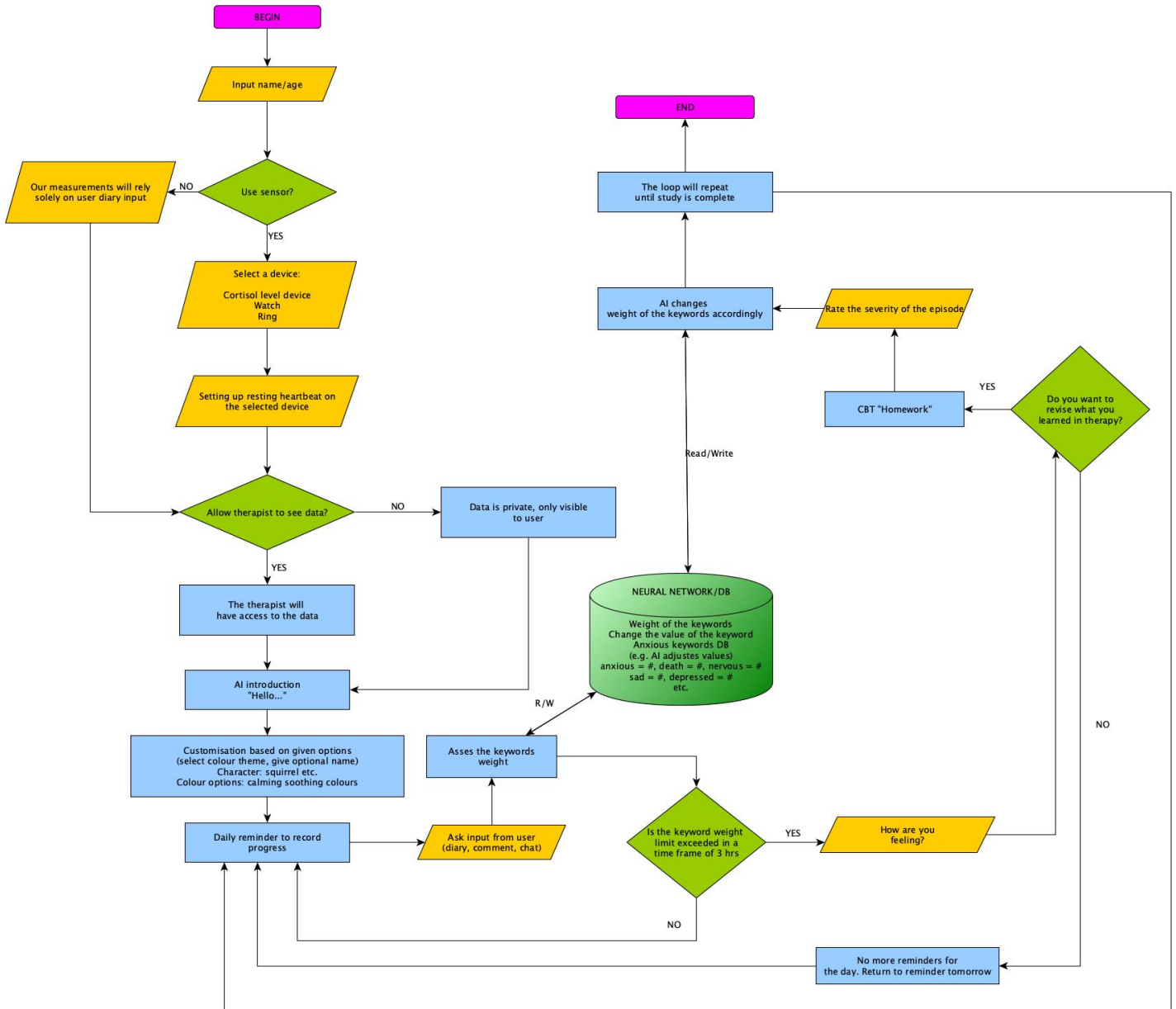
The app might also make the patient feel more like their companion by being friendly to the user and sending them regular reminders regarding how they feel or about their day. The app might send these reminders when a user is in a certain location and/or time of day where the user has indicated they often feel anxious. If, for example, the user often feels anxious at school and the app recognises the user is at their school, the app might send a reminder.

In our effort to make the user feel more at ease while using the app, we have also conducted research to make sure the design of the app is calming. Our findings are that colours with pastel hues and a natural theme could potentially have a calming effect within our app.

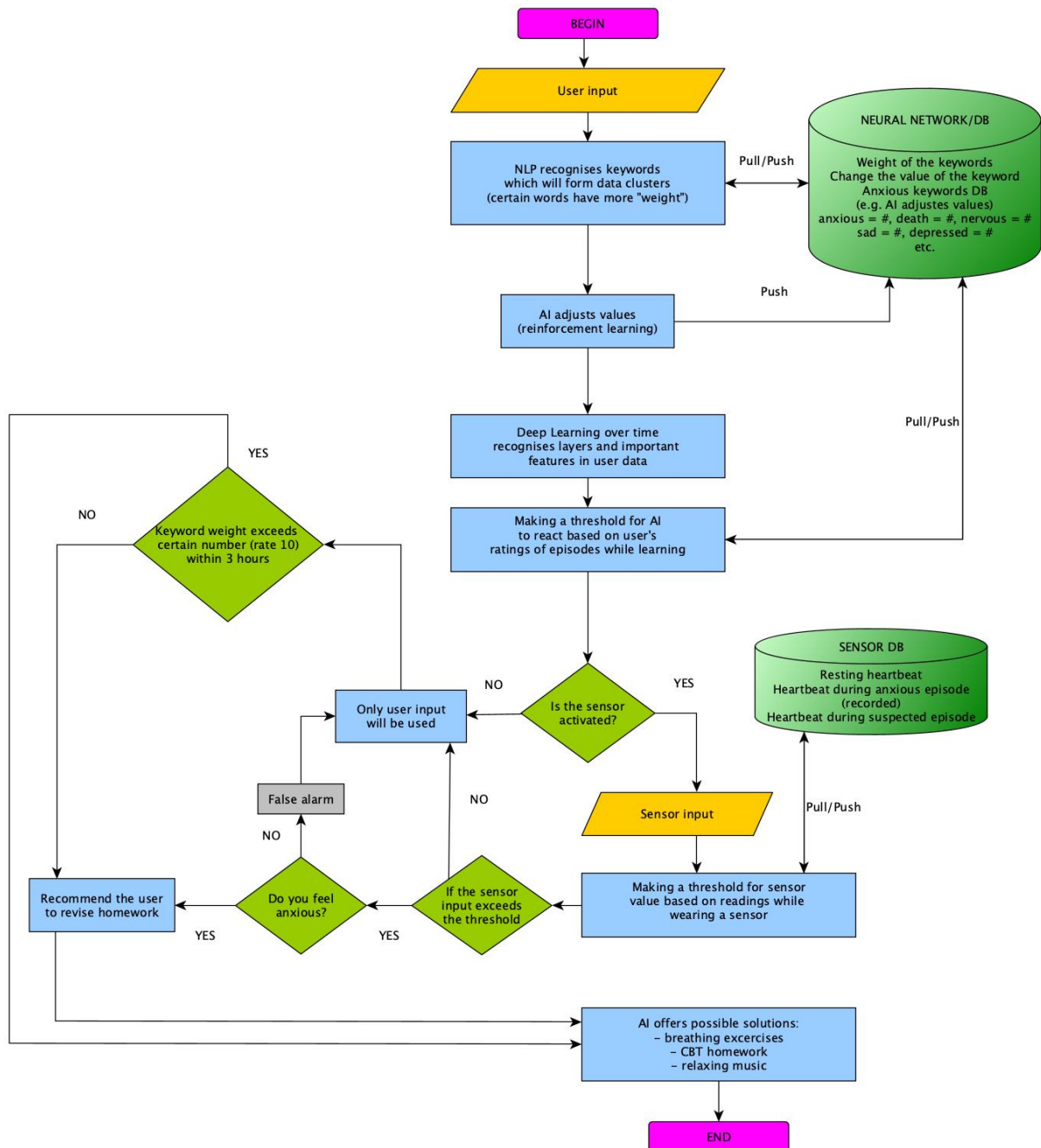
This project has allowed us to combine five different perspectives and sources of knowledge, with each group member enhancing the quality of the research project. We were able to highlight potential improvements in each other's work and provide different viewpoints to further refine the project. Through the presence of a group member with past anxiety troubles, we gained an insight into the details of a patient struggling with anxiety, which allowed us to obtain a deeper understanding of the patient's perspective on the disorder and adapt our research accordingly. The addition of a psychologist helped us verify the validity and professionalism of our methods, while educating us more on the topic of anxiety and interaction with patients. Through our application, we aim to raise awareness of GAD and allow anxiety treatment to become easier to access for patients of all backgrounds around the world. It still forms a barrier in the daily lives of millions of people each year, and we seek to decrease this number. With Willow, we hope that GAD patients will be provided with more guidance in their journey to serenity.

Appendixes

Appendix 1: User experience flowchart



Appendix 2: System flowchart



Appendix 3: Survey

GAD-7

Over the last 2 weeks, how often have you been bothered by the following problems?	Not at all	Several days	More than half the days	Nearly every day
1. Feeling nervous, anxious or on edge	0	1	2	3
2. Not being able to stop or control worrying	0	1	2	3
3. Worrying too much about different things	0	1	2	3
4. Trouble relaxing	0	1	2	3
5. Being so restless that it is hard to sit still	0	1	2	3
6. Becoming easily annoyed or irritable	0	1	2	3
7. Feeling afraid as if something awful might happen	0	1	2	3

Total Score — = Add Columns — + — + —

If you checked off any problems, how difficult have these problems made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

Figure 1. The generalized anxiety disorder 7-item (GAD-7) scale.

Appendix 4: Questions for therapist and psychologist Doris Rafajlovski

- What kind of procedure would a psychologist use to recognise anxiety?
- Which symptoms can be alarming and should issue a warning sign to the therapist?
- What is the easiest way to help an anxious person?
- What is the easiest way to stop anxious thoughts?
- What is the best way to direct ruminating thoughts?
- Which symptom can be treated or cured with an app/personal digital assistant?
- Does she have experience or know anybody who does online therapy?
- Is online therapy as effective as in person? Based on experience during corona?
- Do you believe that an app can be used as an add-on to the therapy, used like a tool?
- Does she think that involving a therapist in the making of the app or behind the app data?

References

- Davis, W. B., & Thaut, M. H. (1989). The Influence of Preferred Relaxing Music on Measures of State Anxiety, Relaxation, and Physiological Responses. *Journal of Music Therapy*, 26(4), 168–187. <https://doi.org/10.1093/jmt/26.4.168>
- Han, H.-R. (2009). Measuring Anxiety in Children: A Methodological Review of Literature. *Asian Nursing Research*, 3(2), 49–62. [https://doi.org/10.1016/S1976-1317\(09\)60016-5](https://doi.org/10.1016/S1976-1317(09)60016-5)
- Henker, B., Whalen, C. K., Jamner, L. D., & Delfino, R. J. (2002). Anxiety, Affect, and Activity in Teenagers: Monitoring Daily Life With Electronic Diaries. *Journal of the American Academy of Child & Adolescent Psychiatry*, 41(6), 660–670. <https://doi.org/10.1097/00004583-200206000-00005>
- March, S. (n.d.). *RUNNING HEAD: ONLINE CBT FOR ADOLESCENT ANXIETY*. 38.
- McLaughlin, K. A., & Nolen-Hoeksema, S. (2011). Rumination as a transdiagnostic factor in depression and anxiety. *Behaviour Research and Therapy*, 49(3), 186–193. <https://doi.org/10.1016/j.brat.2010.12.006>
- Mehroof, M. (n.d.). Online Gaming Addiction: The Role of Sensation Seeking, Self-Control, Neuroticism, Aggression, State Anxiety, and Trait Anxiety. 4.
- Parlak, O., Keene, S. T., Marais, A., Curto, V. F., & Salleo, A. (2018). Molecularly selective nanoporous membrane-based wearable organic electrochemical device for noninvasive cortisol sensing. *Science Advances*, 4(7), eaar2904. <https://doi.org/10.1126/sciadv.aar2904>
- Sawyer, M. C., & Nunez, D. E. (2014). Cognitive-Behavioral Therapy for Anxious Children: From Evidence to Practice. *Worldviews on Evidence-Based Nursing*, 11(1), 65–71. <https://doi.org/10.1111/wvn.12024>
- Song, C. et al., (2019). Physiological and Psychological Effects of Viewing Forests on Young Women. *International Journal of Environmental Research and Public Health*, 16(2). <https://doi.org/10.3390/ijerph16020229>
- Spitzer, R. et al., (2006), A Brief Measure for Assessing Generalized Anxiety Disorder, *Arch Intern Med*. <https://jamanetwork.com/journals/jamainternalmedicine/fullarticle/410326>
- Watkins, E. (2015). Psychological treatment of depressive rumination. *Current Opinion in Psychology*, 4, 32–36. <https://doi.org/10.1016/j.copsyc.2015.01.020>
- Wilms, L., & Oberfeld, D. (2018). Color and emotion: Effects of hue, saturation, and brightness. *Psychological Research*, 82(5), 896–914. <https://doi.org/10.1007/s00426-017-0880-8>
- Wuthrich, V. M., Rapee, R. M., Cunningham, M. J., Lyneham, H. J., Hudson, J. L., & Schniering, C. A. (2012). A Randomized Controlled Trial of the Cool Teens CD-ROM Computerized Program for Adolescent Anxiety. *Journal of the American Academy of Child & Adolescent Psychiatry*, 51(3), 261–270. <https://doi.org/10.1016/j.jaac.2011.12.002>

Websites

- <https://adaa.org> (Anxiety and Depression Association of America), 21/10/2020